

Calibration Where It Counts: Cost- and Data-Informed Isotonic Regression

Gaurav Sood

contact@gsood.com

Abstract

Thresholded decisions turn probability errors into utility losses. Pooling-based monotone calibrators such as isotonic regression are flexible and reliable but can collapse distinct scores into the same probability on wide *plateaus*. We adopt a decision-economic view: choose and tune a calibrator that reduces deployment cost by improving reliability where it affects decisions and by preserving discrimination where it matters (Elkan, 2001; Vickers and Elkin, 2006).

We contribute three pieces. First, a practical *diagnostic suite*—global bootstrap tie stability, within-plateau concordance, minimum detectable difference, and progressive-sampling diversity—with a decision rule that labels plateaus as *supported*, *limited-data*, or *inconclusive*. Second, two adaptive calibrators: **Relaxed PAVA**, which allows bounded local slack and reduces via a cumulative shift to one weighted isotonic projection; and **density-aware smoothed isotonic**, which smooths locally then projects back onto the monotone cone. Third, a formal, convex **cost- and data-informed isotonic** calibrator (CDI-ISO) that encodes economically weighted, variance-aware *local minimum-slope* constraints and solves in $O(n)$ time through a single isotonic pass (Barlow et al., 1972; Robertson et al., 1988). We implement all methods in a scikit-learn-compatible library and evaluate with proper scores and decision curves.

1 Introduction

A probabilistic predictor is *calibrated* when predicted probabilities match empirical frequencies (DeGroot and Fienberg, 1983). Calibration matters because real deployments threshold probabilities: a small bias around the operating point can swing treatment, triage, or approval decisions (Niculescu-Mizil and Caruana, 2005; Jiang et al., 2012). Isotonic regression is a popular post-hoc calibrator because it is flexible and enforces monotonicity (Zadrozny

and Elkan, 2002). Its strength is also a liability: by pooling adjacent violators into constant blocks, isotonic often produces broad *plateaus* that erase within-block discrimination.

Economic lens. Thresholds encode asymmetric costs (Elkan, 2001). Decision-curve analysis (DCA) summarizes utility as net benefit versus threshold, with threshold odds $t/(1-t)$ weighting false positives relative to true positives (Vickers and Elkin, 2006). From this perspective, calibration is economically instrumental: better reliability near the operating point reduces expected loss, while avoidable plateaus can undercut useful ranking.

Aim. We seek to (i) diagnose whether a plateau is *supported* by data or merely *limited* by sample size, and (ii) adapt the calibrator locally when data and operating costs justify recovering discrimination.

1.1 Contributions

Our organizing principle is simple: *reduce decision cost without gratuitously destroying discrimination*.

- **Diagnostics → action.** A practical suite—global bootstrap tie stability, within-plateau concordance (WPC; a Wilcoxon/Mann–Whitney–style concordance inside the tied region), minimum detectable difference (two-proportion power near plateau boundaries), and progressive-sampling diversity—feeds a decision rule that labels plateaus as *supported*, *limited-data*, or *inconclusive*. Each label maps to an action (keep, relax, or smooth).
- **Algorithms with linear-time solvers.** (a) *Relaxed PAVA* allows bounded local slack and reduces to a single weighted isotonic projection via a cumulative shift (inherits $O(n)$ complexity on a total order (Barlow et al., 1972; Robertson et al., 1988)); (b) *density-aware smoothed isotonic* uses local windows, smooths, and reprojects to the monotone cone (Ramsay, 1998; Meyer, 2008; Jiang et al., 2011).
- **Formal cost- and data-informed calibration.** *CDI-ISO* is a convex projection with *local minimum-slope* constraints that are evidence-gated and economically weighted, solved exactly by one isotonic pass. This unifies standard isotone (Zadrozny and Elkan, 2002), relaxed isotone (cf. penalty-based Tibshirani et al., 2011 and ENIR Pakdaman Naeini and Cooper, 2016), and minimum-slope isotone in a single formulation.

2 Background and Related Work

We summarize how calibration is defined and used, how isotonic regression produces plateaus, and how prior work relaxes or smooths monotone fits. We close with decision-aware perspectives and the openings they leave.

2.1 What calibration is and why it is used

A predictor is calibrated if among examples assigned probability p , a fraction p are positive (DeGroot and Fienberg, 1983). Calibration matters because thresholding converts probability error into decision losses (Niculescu-Mizil and Caruana, 2005; Jiang et al., 2012). Post-hoc methods fall into two families. *Parametric* approaches—Platt/logistic scaling (Platt, 1999), beta calibration (Kull et al., 2017), and temperature/matrix/vector scaling for deep nets (Guo et al., 2017)—fit low-parameter transformations. *Nonparametric* approaches—histogram binning (Zadrozny and Elkan, 2001) and isotonic regression (Zadrozny and Elkan, 2002)—trade smoothness for flexibility and monotonicity.

2.2 Isotonic regression and the origin of plateaus

Univariate isotonic fits a nondecreasing sequence by minimizing weighted squared error subject to order constraints. The Pool Adjacent Violators Algorithm (PAVA) solves this in $O(n)$ time on a total order (Ayer et al., 1955; Barlow et al., 1972; Robertson et al., 1988). PAVA’s solution is a right-continuous, nondecreasing *step function*: adjacent violators are merged into contiguous *blocks* and each block is assigned its weighted mean, so all scores in a block receive the same $\hat{p}$ (Barlow et al., 1972; Robertson et al., 1988). These constant blocks are the *plateaus*. They can reflect true flatness in the underlying calibration function, or they can be artifacts of sparsity and noise that force pooling in finite samples; both behaviors are observed in practice (Zadrozny and Elkan, 2002).

2.3 Relaxing or smoothing monotone fits

Two strands try to mitigate over-coarse plateaus while retaining monotonicity. **Relaxations** penalize violations instead of forbidding them. Nearly-isotonic regression tunes a penalty to trade bias and variance (Tibshirani et al., 2011); ENIR calibrates by ensembling near-isotonic fits (Pakdaman Naeini and Cooper, 2016). **Smooth monotone models** impose smoothness alongside monotonicity via basis functions and constraints (Ramsay, 1998; Meyer, 2008); in calibration, “smooth isotonic” tempers stepwise fits with local smoothing before reimposing

monotonicity (Jiang et al., 2011). These approaches choose *how much* to relax or smooth, but not *where* it is justified by data or *why* it matters economically.

2.4 Decision-aware perspectives

Cost-sensitive learning formalizes how false-positive/false-negative costs move the operating threshold (Elkan, 2001). DCA summarizes *net benefit* versus threshold using the odds $t/(1-t)$ to weight false positives (Vickers and Elkin, 2006). These tools clarify *which* thresholds matter, but they do not change the calibration mapping itself.

Openings. We identify three gaps: (i) practical diagnostics that separate genuine from data-limited plateaus; (ii) *local*, evidence-based mechanisms that break ties only where supported; and (iii) integration of decision costs into the calibration mapping, so recovering discrimination is done where it improves utility.

3 Problem Setup and Decision-Economic Framework

Let s denote model scores, $g(s) \in [0, 1]$ the calibrator, and $\pi(s) = \mathbb{P}(y=1 \mid s)$ the (unknown) truth. Thresholding at t yields a decision $a \in \{0, 1\}$. DCA defines net benefit

$$\text{NB}(t) = \text{TPR}(t) \pi - \text{FPR}(t) (1 - \pi) \frac{t}{1-t},$$

where $\frac{t}{1-t}$ are threshold odds (Vickers and Elkin, 2006). We report NB curves to connect calibration to utility.

For model selection we consider

$$\mathcal{J}(g; \lambda, \mathcal{T}) = \mathbb{E}[S(y, g(s))] + \lambda \mathbb{E}_{t \sim \mathcal{T}}[\text{DiscLoss}_t(g)],$$

with S a proper score (Brier or NLL) (Gneiting and Raftery, 2007). The term DiscLoss_t penalizes ties/ranking loss localized near threshold t (e.g., $1 - \text{WPC}$ inside a plateau that covers t). We use $\mathcal{J}$ to guide hyperparameters; we do not claim Bayes-risk optimality.

4 Plateau Diagnostics

Let (s_i, y_i) be calibration data sorted by s_i and $\hat{p}_i = g(s_i)$ the fitted probabilities.

Definition 1 (Plateau) A plateau is a maximal index interval $P = [i_s, i_e]$ with $\hat{p}_{i_s} = \dots = \hat{p}_{i_e}$. We summarize P by its score span $[s_{\min}, s_{\max}]$, size $|P|$, and label variance.

Global bootstrap tie stability. Resample the *entire* calibration set (optionally stratified by score quantiles), refit $g^{(b)}$, and evaluate $g^{(b)}$ at the observed scores in $[s_{\min}, s_{\max}]$. Stability τ is the fraction of bootstraps with empirical range $< \epsilon$ on that span; stable plateaus persist under global refits.

Within-plateau concordance (WPC). With P_+ and P_- the positives/negatives in P ,

$$\text{WPC}_P = \frac{1}{|P_+||P_-|} \sum_{i \in P_+} \sum_{j \in P_-} \mathbf{1}\{s_i > s_j\},$$

with a Wilcoxon/Mann–Whitney test against 0.5 (ties scored 0.5) to detect residual ranking.

Minimum detectable difference (MDD). At P ’s boundaries, a two-proportion power calculation estimates the minimum $\Delta = \mu_1 - \mu_0$ detectable at level α ; large MDD implies low power to distinguish flatness from slope.

Progressive-sampling diversity. Fit isotonic on subsamples of size n and track a tie metric (or unique-value ratio). Increasing curves suggest additional data would reduce plateaus.

Decision rule. *Supported:* high τ , $\text{WPC} \approx 0.5$, small MDD. *Limited-data:* low τ or WPC far from 0.5 or large MDD. *Inconclusive:* otherwise. Thresholds are tuned on held-out validation to avoid optimism.

5 Methods

5.1 Relaxed PAVA: local slack with $O(n)$ reduction

Standard PAVA solves $\min_{z_1 \leq \dots \leq z_n} \sum_i w_i (y_i - z_i)^2$ in $O(n)$ time on a total order (Ayer et al., 1955; Barlow et al., 1972; Robertson et al., 1988). We allow bounded local violations by imposing adjacent-difference lower bounds

$$z_{i+1} - z_i \geq L_i, \quad L_i \leq 0.$$

Data-adaptive L_i can come from adjacent *block-mean* differences (percentile rule) or from a variance-aware bound

$$L_i = -z_\alpha \sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_i} + \frac{1}{n_{i+1}} \right)},$$

where $\hat{p}$ is the pooled rate of neighboring blocks and n_i their sizes. A cumulative shift $R_{i+1} = R_i + L_i$ reduces the problem to a single weighted isotonic projection on $y'_i = y_i - R_i$, with solution $z_i^* = u_i^* + R_i$; clipping to $[0, 1]$ preserves monotonicity because clipping is non-decreasing (Barlow et al., 1972; Robertson et al., 1988).

5.2 CDI-ISO: cost- and data-informed isotonic

We encode *economically weighted, variance-aware minimum slope* near operating thresholds and allow variance-aware relaxation elsewhere. Let $\Delta_i = z_{i+1} - z_i$. We solve the convex projection

$$\min_{z \in \mathbb{R}^n} \sum_{i=1}^n w_i (y_i - z_i)^2 \quad \text{s.t.} \quad \Delta_i \geq L_i \quad (i = 1, \dots, n-1), \quad (1)$$

with $L_i = \phi_i - \varepsilon_i$. Here $\phi_i \geq 0$ is an economically weighted minimum slope near thresholds; $\varepsilon_i \geq 0$ is a variance-aware relaxation elsewhere.

Constructing L_i . Let $\bar{s}_i = (s_i + s_{i+1})/2$ and define economics weights

$$w_i^{\text{econ}} = \mathbb{E}_{t \sim \mathcal{T}} [K_h(|\bar{s}_i - t|)],$$

with triangular kernel K_h (half-width h) concentrating mass near thresholds of interest (from cost ratios or DCA Elkan, 2001; Vickers and Elkin, 2006). For data evidence, compute a lower confidence bound for adjacent block differences using the pooled standard error:

$$\Delta_i^{\text{LCB}} = (\hat{p}_{i+1} - \hat{p}_i) - z_\alpha \sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_i} + \frac{1}{n_{i+1}} \right)}.$$

Define

$$\phi_i = \gamma w_i^{\text{econ}} [\Delta_i^{\text{LCB}}]_+, \quad \varepsilon_i = (1 - w_i^{\text{econ}}) z_\alpha \sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_i} + \frac{1}{n_{i+1}} \right)},$$

with $\gamma \in [0, 1]$ setting a global slope budget; if $\Delta_i^{\text{LCB}} \leq 0$ we set $\phi_i = 0$ (no enforced slope without evidence).

Linear-time solver. Let $R_1=0$, $R_{i+1}=R_i+L_i$, and set $u_i = z_i - R_i$, $y'_i = y_i - R_i$. Then (1) reduces to

$$u^* = \arg \min_{u_1 \leq \dots \leq u_n} \sum_i w_i (y'_i - u_i)^2, \quad z_i^* = u_i^* + R_i,$$

which is a single weighted isotonic projection solvable in $O(n)$ time on a total order (Barlow et al., 1972; Robertson et al., 1988). Clipping z^* to $[0, 1]$ preserves monotonicity.

5.3 Density-aware smoothed isotonic

We first smooth locally, then project to the monotone cone: (1) choose a kNN/quantile window W_i around s_i ; (2) apply local regression to obtain $\tilde{y}_i$ (clip to $[0, 1]$); (3) run a monotone projection (PAVA) on $\tilde{y}$ to obtain $\hat{y}^{\text{smooth}} = \Pi_{\text{mono}}(\tilde{y})$ (Ramsay, 1998; Meyer, 2008; Jiang et al., 2011). CDI-ISO can replace the final projection when minimum-slope constraints are desired.

6 Theory: scope and guarantees

Feasibility and complexity. Weighted PAVA on a total order is $O(n)$ (Barlow et al., 1972; Robertson et al., 1988). The shift-to-PAVA reduction preserves this complexity and ensures feasibility for any real L_i ; clipping to $[0, 1]$ preserves monotonicity.

Power near plateaus. Let boundary means be μ_0, μ_1 with $\Delta = \mu_1 - \mu_0$. A Hoeffding-style back-of-the-envelope suggests that detecting $\Delta > 0$ with error $\leq \delta$ requires local effective sample size $O(\log(1/\delta)/\Delta^2)$. This is a heuristic for interpreting diagnostic power, not a minimax rate.

7 Conclusion

Economics-guided diagnostics and CDI-ISO help practitioners keep calibration honest without sacrificing discrimination where it affects decisions. By encoding evidence- and cost-informed local constraints in a convex projection with a linear-time solver, we make it practical to calibrate where it counts.

References

- Ayer, M., Brunk, H. D., Ewing, G. M., Reid, W. T., and Silverman, E. (1955). An empirical distribution function for sampling with incomplete information. *The Annals of Mathematical Statistics*, 26(4):641–647.
- Barlow, R. E., Bartholomew, D. J., Bremner, J. M., and Brunk, H. D. (1972). *Statistical Inference Under Order Restrictions: The Theory and Application of Isotonic Regression*. John Wiley & Sons, New York.
- DeGroot, M. H. and Fienberg, S. E. (1983). The comparison and evaluation of forecasters. *Journal of the Royal Statistical Society: Series D (The Statistician)*, 32(1–2):12–22.
- Elkan, C. (2001). The foundations of cost-sensitive learning. In *Proceedings of the 17th International Joint Conference on Artificial Intelligence (IJCAI)*, pages 973–978.
- Gneiting, T. and Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378.
- Guo, C., Pleiss, G., Sun, Y., and Weinberger, K. Q. (2017). On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, volume 70, pages 1321–1330. PMLR.
- Jiang, X., Osl, M., Kim, J., and Ohno-Machado, L. (2012). Calibrating predictive model estimates to support personalized medicine. *Journal of the American Medical Informatics Association*, 19(2):263–274.
- Jiang, X., Osl, M., and Ohno-Machado, L. (2011). Smooth isotonic regression: A new method to calibrate predictive models. *Journal of the American Medical Informatics Association*, 18(3):277–282.
- Kull, M., Silva Filho, T. M., and Flach, P. (2017). Beta calibration: A well-founded and easily implemented improvement on logistic calibration for binary classifiers. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 54, pages 623–631. PMLR.
- Meyer, M. C. (2008). Inference using shape-restricted regression splines. *The Annals of Applied Statistics*, 2(3):1013–1033.
- Niculescu-Mizil, A. and Caruana, R. (2005). Predicting good probabilities with supervised learning. In *Proceedings of the 22nd International Conference on Machine Learning (ICML)*, pages 625–632. ACM.

- Pakdaman Naeini, M. and Cooper, G. F. (2016). Binary classifier calibration using an ensemble of near isotonic regression models. In *Proceedings of the IEEE International Conference on Data Mining (ICDM)*, pages 360–369.
- Platt, J. (1999). Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In Smola, A. J., Bartlett, P., Schölkopf, B., and Schuurmans, D., editors, *Advances in Large Margin Classifiers*, pages 61–74. MIT Press.
- Ramsay, J. O. (1998). Estimating smooth monotone functions. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 60(2):365–375.
- Robertson, T., Wright, F. T., and Dykstra, R. L. (1988). *Order Restricted Statistical Inference*. John Wiley & Sons, Chichester, UK.
- Tibshirani, R. J., Höfling, H., and Tibshirani, R. (2011). Nearly-isotonic regression. *Technometrics*, 53(1):54–61.
- Vickers, A. J. and Elkin, E. B. (2006). Decision curve analysis: A novel method for evaluating prediction models. *Medical Decision Making*, 26(6):565–574.
- Zadrozny, B. and Elkan, C. (2001). Obtaining calibrated probability estimates from decision trees and naive Bayesian classifiers. In *Proceedings of the 18th International Conference on Machine Learning (ICML)*, pages 609–616. Morgan Kaufmann.
- Zadrozny, B. and Elkan, C. (2002). Transforming classifier scores into accurate multiclass probability estimates. In *Proceedings of the 8th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 694–699. ACM.

A Optional note on binomial intervals

For constructing Δ_i^{LCB} one can replace the normal-approximation SE with Wilson-style intervals, which have better finite-sample behavior; see Brown, Cai, and DasGupta (2001). If you prefer to avoid extra dependencies, the normal approximation used in the main text is consistent with standard practice.